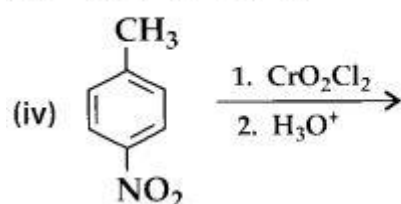
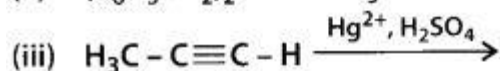
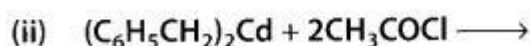
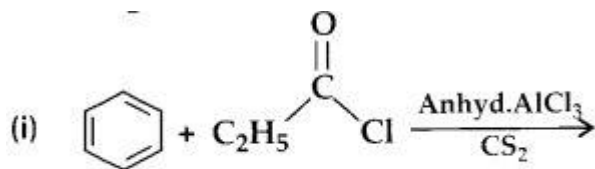


WORKSHEET ON ALDEHYDES, KETONES AND

CARBOXYLIC ACID

1. Write the structures of the products of the following reactions:



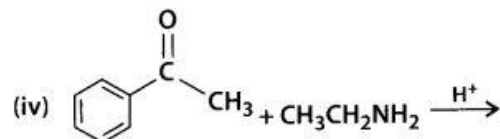
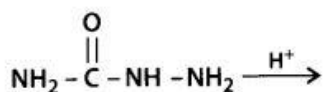
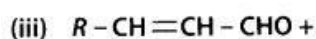
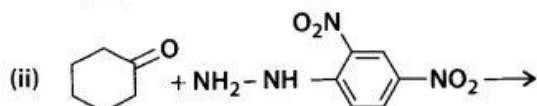
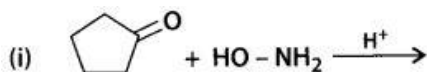
2. Arrange the following compounds in increasing order of their boiling points CH_3CHO , $\text{CH}_3\text{CH}_2\text{OH}$, CH_3OCH_3 , $\text{CH}_3\text{CH}_2\text{CH}_3$

3. Arrange the following compounds in increasing order of their reactivity in nucleophilic addition reactions.

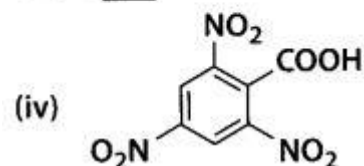
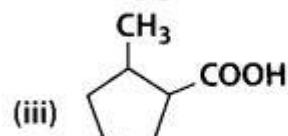
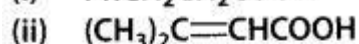
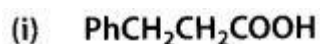
a) Ethanal, Propanal, Propanone, Butanone.

b) Benzaldehyde, p-Tolualdehyde, p-Nitrobenzaldehyde, Acetophenone.

4. Predict the products of the following reactions:



5. Give the IUPAC names of the following compounds:



6. Show how each of the following compounds can be converted to benzoic acid.

- a) Ethylbenzene
- b) Acetophenone
- c) Bromobenzene
- d) Phenylethene (Styrene)

7. Which acid of each pair shown here would you expect to be stronger?



8. Predict the products formed when cyclohexane carbaldehyde reacts with following reagents.

- a) PhMgBr and then H_3O^+
- b) Tollens' reagent
- c) Semicarbazide and weak acid
- d) Excess ethanol and acid
- e) Zinc amalgam and dilute hydrochloric acid

9. How will you convert ethanal into the following compounds?

- a) Butane-1,3-diol
- b) But-2-enal
- c) But-2-enoic acid

10. Write structural formulas and names of four possible aldol condensation products from propanal and butanal. In each case, indicate which aldehyde acts as nucleophile and which as an electrophile

11. An organic compound with the molecular formula $C_9H_{10}O$ forms 2,4-DNP derivative, reduces Tollens' reagent and undergoes Cannizzaro reaction. On vigorous oxidation, it gives 1, 2-benzenedicarboxylic acid. Identify the compound.

12. An organic compound (A) (molecular formula $C_8H_{16}O_2$) was hydrolysed with dilute sulphuric acid to give a carboxylic acid (B) and an alcohol (C). Oxidation of (C) with chromic acid produced (8). (C) on dehydration gives but-1-ene. Write equations for the reactions involved.

13. Give simple chemical tests to distinguish between the following pairs of compounds.

- a) Propanal and Propanone
- b) Acetophenone and Benzophenone
- c) Phenol and Benzoic acid
- d) Benzoic acid and Ethyl benzoate
- e) Pentan-2-one and Pentan-3-one
- f) Benzaldehyde and Acetophenone
- g) Ethanal and Propanal

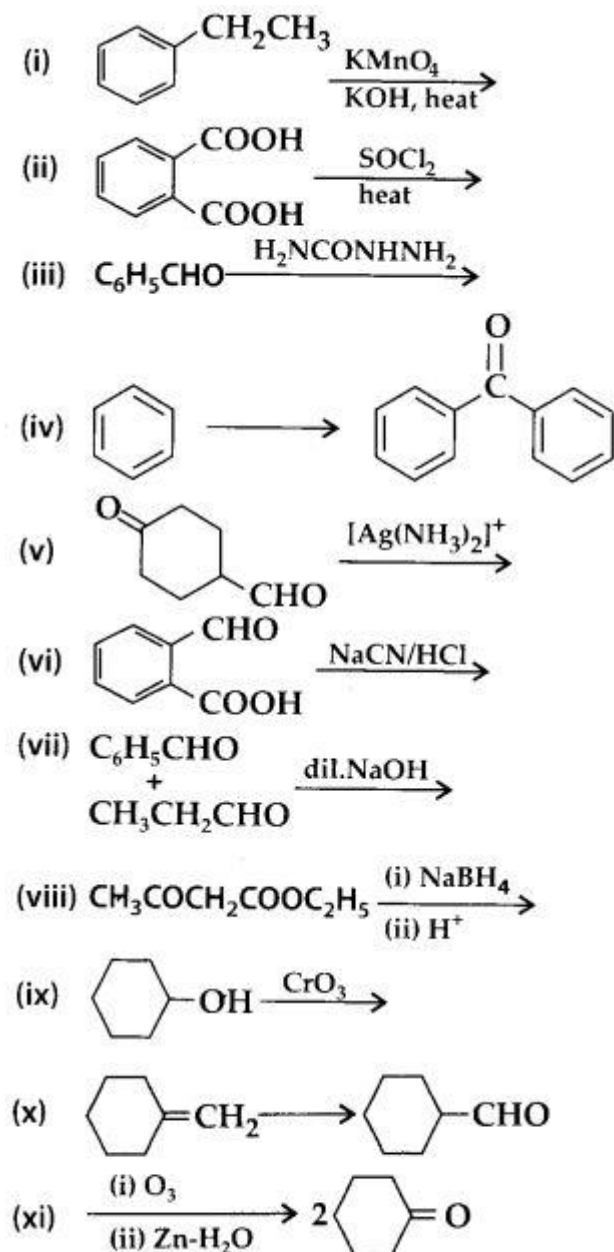
14. How will you prepare the following compounds from benzene? You may use any inorganic reagent and any organic reagent having not more than one carbon atom

- a) Methyl benzoate
- b) m-Nitrobenzoic acid
- c) p-Nitrobenzoic acid
- d) Phenylacetic acid
- e) p-Nitrobenzaldehyde

15. How will you bring about the following conversions in not more than two steps?

- a) Propanone to Propene
- b) Benzoic acid to Benzaldehyde
- c) Ethanol to 3-Hydroxybutanal
- d) Benzene to m-Nitroacetophenone
- e) Benzaldehyde to Benzophenone
- f) Bromobenzene to 1-Phenylethanol
- g) Benzaldehyde to 3-Phenylpropan-1-ol
- h) Benzaldehyde to α -Hydroxyphenylacetic acid
- i) Benzoic acid to m-Nitrobenzyl alcohol

16. Complete each synthesis by giving missing starting material, reagents or products



17. Give possible explanation for each of the following:

- Cyclohexanone forms cyanohydrin in good yield but 2, 2,6 trimethylcyclohexa- none does not.
- There are two – NH₂ groups in semicarbazide. However, only one is involved in the formation of semicarbazones.
- During the preparation of esters from a carboxylic acid and an alcohol in the presence of an acid catalyst, the water or the ester should be removed as soon as it is formed.